# Fiber Optic Communication Systems



# Introduction

- Title:  Fiber Optic Communication Systems
- Credit Hours: 3+1
- Semester: 7th
- Course Teacher: Professor Dr. Monir Morshed

- **Recommended Book:**

  *Fiber Optic Communications by Joseph C. Palais, Latest Edition.*

- **Reference Books:**

  *Optical Fiber Communications: Principles and Practice by John M. Senior, Latest Edition, Prentice Hall.*

  *Optical Fiber Communications by Gerd Keiser, Latest Edition, McGraw Hill.*

  *Fiber-Optic Communication Systems by Govind P. Agrawal*

# Objective

- To realize the significance of optical fiber communications.
- To understand the construction and characteristics of optical fiber cable.
- To develop the knowledge of optical signal sources and power launching.
- To identify and understand the operation of various optical detectors.
- To under the design of optical systems and WDM.

# Course Contents

- Overview
  - Historical perspectives
  - The basic communication systems
  - Nature of light
  - Advantages of light
  - Application of fiber optic communications
- Optics Review
  - Ray theory and applications
  - Lenses
  - Imaging
  - Numerical aperture
  - Diffraction

## Course Contents cont…

- Lightwave Fundamentals
  - Electromagnetic waves
  - Dispersion, pulse dispersion and information
  - Polarization
  - Critical angle Reflections
- Optic Fiber Waveguide
  - Step-index Fiber
  - Graded-index fiber
  - Attenuation
  - Pulse distortion and information rate in optic fiber
  - Construction of optic fiber
  - Optic fiber cables

## Course Contents cont...

- Light Sources
  - Light emitting diodes
  - Light emitting diodes operating characteristics
  - Laser principles
  - Laser diodes
  - Laser diodes operating characteristics
  - Distributed Feed-back Laser diodes
  - Optical Amplifiers
  - Vertical-Cavity Surface-Emitting Laser (VCSEL) Diodes
- Light Detectors
  - Principles of photo-detection
  - Semiconductor photo diodes
  - PIN photodiodes
  - Avalanche Photodiodes

## Course Contents cont…

- Couplers and Connectors
  - Connector Principles
  - Splices
  - Connectors
  - Source Coupling
- Distributed Networks and fiber components
- FDDI, SONET, ATM,SDH,WDM
  - Fiber Bragg Gratings
  - Other components
- PON
- Noise and Detection
  - Thermal and shot noise
  - Signal to noise ratio
  - Error rates
  - Mode-noise, Mode partition noise, Amplifier noise, Laser noise and jitter
  - Additional noise contributor
  - Receiver circuit design

# Course Contents cont…

- Systems Design
  - Analog Systems design
  - Digital systems design
- DWDM Filters
  - Fiber Grating Filters
  - Fabry Perot Filters
  - Multilayer Dielectric Thin-Film Filters
  - Mach-Zehnder Interferometers

**Photonic Networks**

# Tentative Grading Policy

| | |
|---|---|
| Attendance | 10% |
| CT | 20% |
| Final Paper | 70% |
| Lab Report | 20% |
| Lab Attendance | 10% |
| Quiz | 20% |
| Lab Final | 50% |

# Why is this course useful to you?



✓Perhaps the most obvious answer is to consider the social impact of optical communications.

✓ In 1966, two engineers at Standard Telecommunication Laboratories in England wrote a paper which in essence started the race to develop optical fiber for communications:

# Why is this course useful to you?

"It is foreseeable that glasses with a bulk loss of about 20 dB/km at around 0.6 micron will be obtained, as the iron-impurity concentration may be reduced to 1 part per million."



✓One year later, the Philco-Ford company tried to predict the future; they were in many ways very close to what we have. (https://www.youtube.com/watch?v=rpq5ZmANp0k

# What happens in an 2021 INTERNET MINITUE?




✓All of this is made possible by a global infrastructure of optical cables

| | SEA-ME-WE | SEA-ME-WE 2 | SEA-ME-WE 3 | SEA-ME-WE 4 | SEA-ME-WE 5 |
|---|---|---|---|---|---|
| Commissioned | June 1985 (Decommissioned June 1999) | October 1994 (Decommissioned October 2006) | August 1999 | November 2005 | November 2016 (expected) |
| Capacity/Length | 12/25MHz 13,500km | 2x560Mbps 18,000km | 8x2.5Gbps 39,000km | 2x64x10Gbps 20,000km | 3x80x100Gbps 20,000km |
| No. of Owners | 22 | 52 | 92 | 16 | 16 |
| Technology | Analog/Copper | PDH/Optical | SDH/WDM/Optical | SDH/DWDM/Optical | Coherent QPSK/Optical |

# Numbers

Some basic Units that may appear frequently in this course are listed in following table for convenient reference

| Unit | Symbol | Measure |
|---|---|---|
| meter | m | length |
| kilogram | kg | mass |
| second | s | time |
| coulomb | C | charge |
| joule | J | energy |
| watt | W | power |
| hertz | Hz | frequency |
| newton | N | force |
| ampere | A | current |
| kelvin | K | temperature |
| degree Celsius | °C | temperature |
| farad | F | capacitance |
| ohm | Ω | resistance |
| volt | V | voltage |
| radian | r | angle |

# Constants & Prefixes

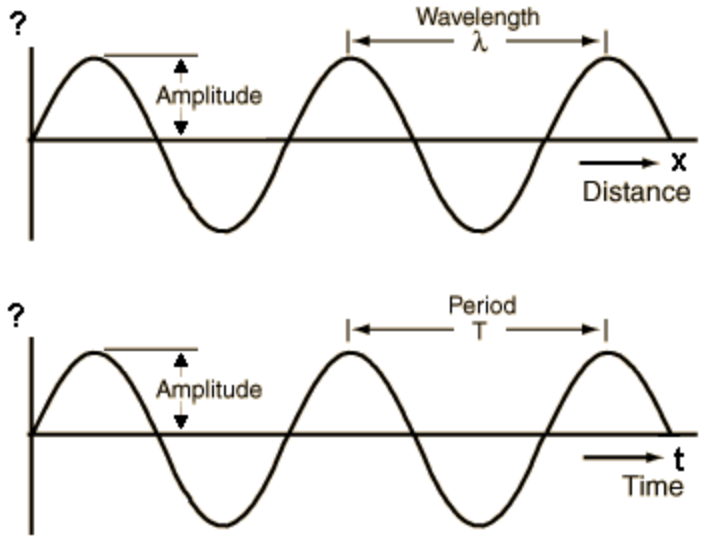
A few important constants and prefixes that will need to study our course.

| Description | Value | Symbol |
|---|---|---|
| Velocity of light | $3 \times 10^{8}$ m/s | $c$ |
| Planck constant | $6.66 \times 10^{-34}$ J × s | $h$ |
| Electron charge | $-1.6 \times 10^{-19}$ C | $-e$ |
| Boltzmann constant | $1.38 \times 10^{-23}$ J/K | $k$ |

| Prefix | Symbol | Multiplication Factor |
|---|---|---|
| tera | T | $10^{12}$ |
| giga | G | $10^{9}$ |
| mega | M | $10^{6}$ |
| kilo | k | $10^{3}$ |
| centi | c | $10^{-2}$ |
| milli | m | $10^{-3}$ |
| micro | $\mu$ | $10^{-6}$ |
| nano | n | $10^{-9}$ |
| pico | p | $10^{-12}$ |
| femto | f | $10^{-15}$ |

# Frequency & Period

- The frequency unit, the hertz, is equivalent to one cycle of oscillation per second .
- The time between successive peaks of an oscillation is called the period.
- If *f* is the wave frequency and *T* is its period then T=1/*f*.




$$Wave\ Period = \frac{1}{Frequency} \quad or \quad T = \frac{1}{f}$$

$$Wave\ Period = \frac{Wavelength}{velocity} \quad or \quad T = \frac{\lambda}{v}$$

*Unit of Wave Period: s*

# Bandwidth requirement for Common Analog Systems

- Following Table, summarizes the bandwidth requirements of several analog systems.

| Message Type | Bandwidth | Comments |
|---|---|---|
| Voice | 4 kHz | single telephone channel |
| Music | 10 kHz | AM radio broadcasting station |
| Music | 200 kHz | FM radio broadcasting station |
| Television | 6 MHz | television broadcasting station |

**Bandwidth**

| Communications | Bandwidth |
|---|---|
| Dial-up (modem/telephone line) | 56Kbps |
| Broadband (ADSL) | up to 20Mbps |
| Wireless (WiFi) | 54Mbps |
| LAN (wired) | 100Mbps (1Gbps) |
| Leased Line (DSL) | up to 1Gbps |

## Bandwidth:

- Bandwidth is a maximum transmission rate of a communications channel.
- Measure in bits per second (bps)
- Kbps, Mbps, Gbps
- A measure of the capacity of communication channel

# Sampling

When analog signals are transmitted digitally, the bit rate depends on the

- rate at which the analog signal is sampled
- the coding scheme.

**Why the standard 4-kHz telephone channel is sampled 8000 times a second?**

- To transmit the signal accurately, the signal should be sampled at a rate of at least twice the highest frequency contained in that signal.
- For this reason the standard 4-kHz telephone channel is sampled 8000 times a second.
- The coding procedure uses 8 bit to present amplitude of each sample
- So a single telephone message needs 64,000 bps.

# Data Rates of the Telephone Systems in the USA

Digital Transmission Rates of the Telephone System in the United States

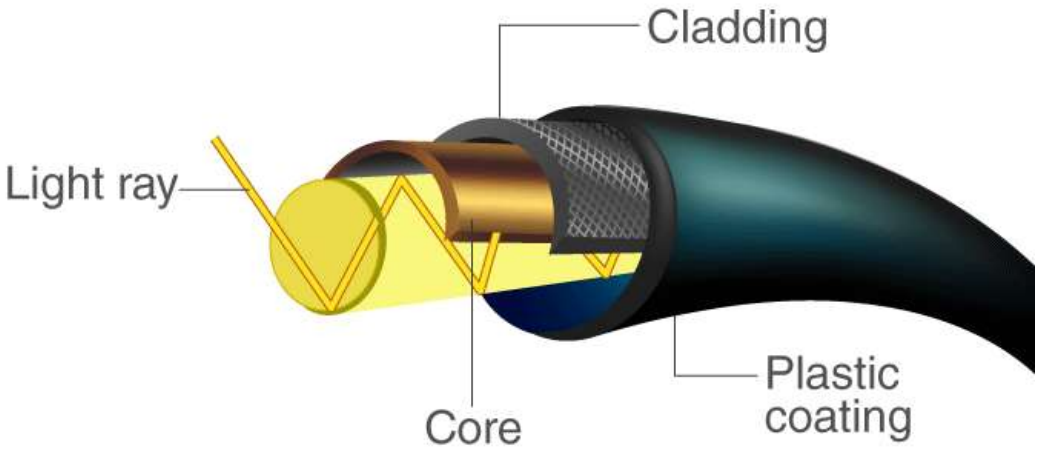
| Number of Voice Channels | Transmission Designation | Signaling Designation | Data Rate |
|---|---|---|---|
| 1 | | | 64 kbps |
| 24 | T1 | DS-1 | 1.5444 Mbps |
| 48 (2 T1 systems) | T1C | DS-1C | 3.152 Mbps |
| 96 (4 T1 systems) | T2 | DS-2 | 6.312 Mbps |
| 672 (7 T2 systems) | T3 | DS-3 | 44.736 Mbps |
| 1344 (2 T3 systems) | T3C | DS-3C | 91.053 Mbps |
| 4032 (6 T3 systems) | T4 | DS-4 | 274.175 Mbps |
| 6048 (9 T3 systems) | — | — | 405 Mbps |
| 8064 (12 T3 systems) | — | — | 565 Mbps |

- A single voice channel needs 64 Kbps.
- A T1 line can carry 24 voice channels or digital data rate of 1.544 Mbps.
- Traditional T1 line use 2 twisted pair copper wires.
- A T1 line can transmit data at a speed of 1.544 Mbps

# Introduction

Fiber-optic communication:

- It is a method of transmitting information from one place to another by sending pulses of light through transparent waveguides.
- The transparent waveguides are made of plastic or glass.
- An optical fiber is a very thin strand of plastic or glass that is used to transmit messages via light
- The light forms an electromagnetic carrier wave that is modulated to carry information.



# Introduction

## Why optical fiber is preferred :

- High bandwidth
- Long distance
- Immunity to electromagnetic interference required
- It can transmit voice, video, and telemetry through LAN, computer networks, or across long distances.

## Application of optical fiber:

- Medical: imaging tools, as laser for surgeries
- Defense/Government: Used as hydrophones for seismic waves and SONAR
- Data storage
- Telecommunications
- Networking
- Broadcast/CATV

# Historical Development

First generation (Graded-index fibers):

- Year implemented: 1980
- Bit rate: 45 Mb/s
- Repeater spacing: 10 km
- Operating wavelength: 0.8 um
- Semiconductor: GaAs

Second generation (Single-mode fibers):

- Year implemented: 1985
- Bit rate: 100 Mb/s to 1.7 Gb/s
- Repeater spacing: 50 km
- Operating wavelength: 1.3 um
- Semiconductor: In GaAsP

# Historical Development

Third generation (Single-mode lasers):

- Year implemented: 1990
- Bit rate: 10 Gb/s
- Repeater spacing: 100 km
- Operating wavelength: 1.55 um

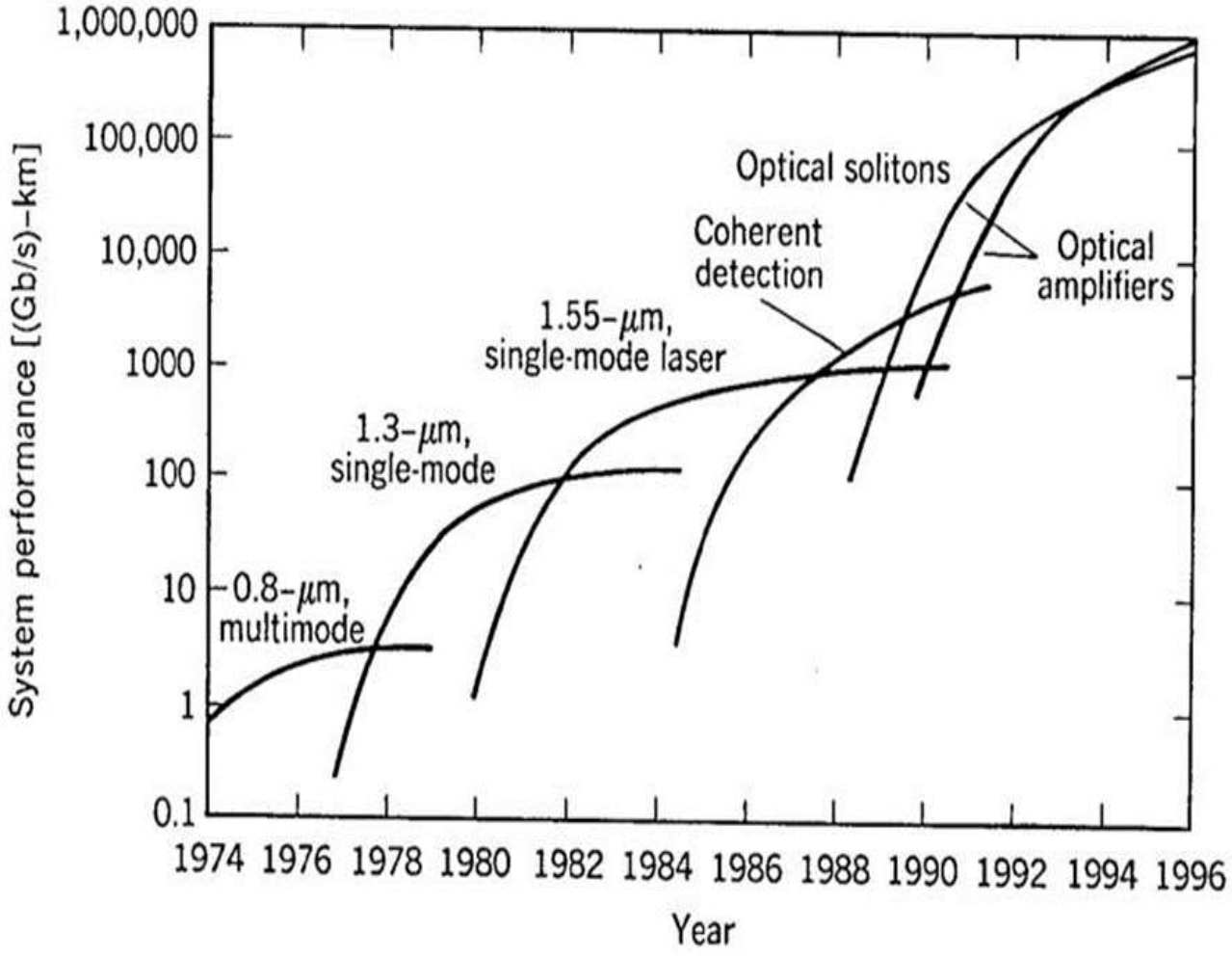
Fourth generation (Optical amplifiers):

- Year implemented: 1996
- Bit rate: 10 Tb/s
- Repeater spacing: > 10,000 km
- Operating wavelength: 1.45 um to 1.62 um

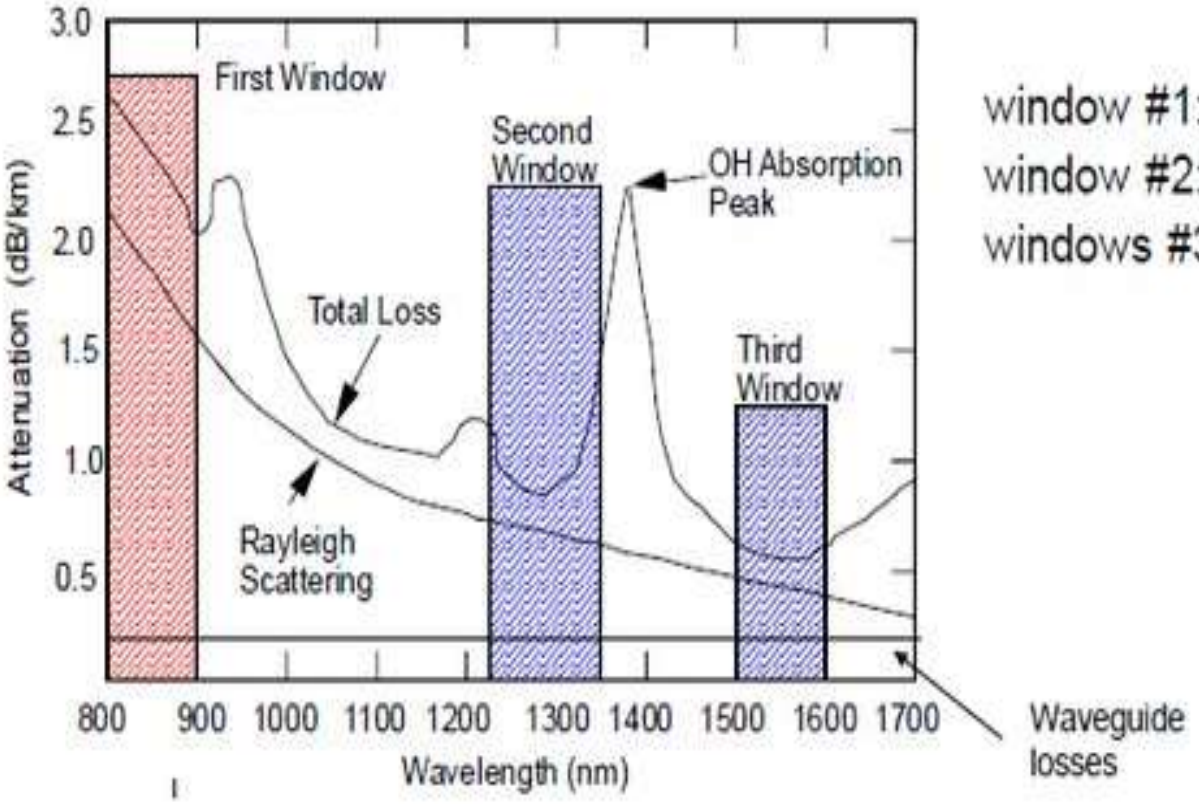
Fifth generation (Raman amplification):

- - Year implemented: 2002
- - Bit rate: 40 Gb/s to 160 Gb/s
- - Repeater spacing: 24,000 km to 35,000 km
- - Operating wavelength: 1.53 um to 1.57 um

# Progress in Lightwave Communication Technology



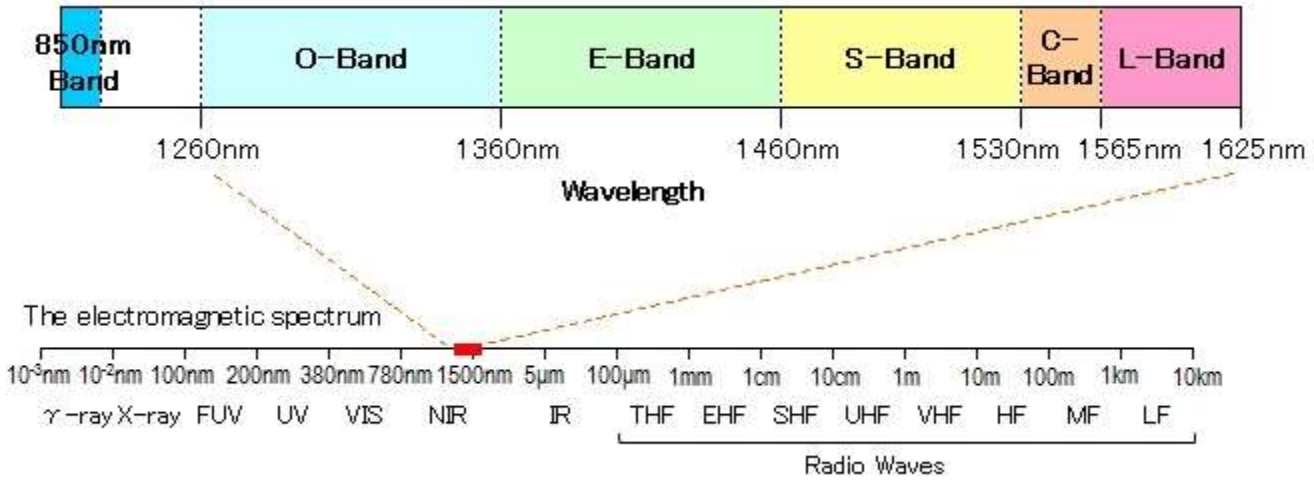
# Transmission loss of silica fiber and optical communication wavelength bands




window #1: 0.8-0.9μm
window #2: 1.25-1.35μm
windows #3: 1.5-1.6 μm

# Electromagnetic spectrum & optical communication wavelength bands

| Band | Description | Wavelength range |
|---|---|---|
| O band | original | 1260–1360 nm |
| E band | extended | 1360–1460 nm |
| S band | short wavelengths | 1460–1530 nm |
| C band | conventional ("erbium window") | 1530–1565 nm |
| L band | long wavelengths | 1565–1625 nm |
| U band | ultralong wavelengths | 1625–1675 nm |



# General OFC System

Basic block diagram of optical fiber communication system consists of following important blocks:

- Transmitter
- Information channel
- Receiver

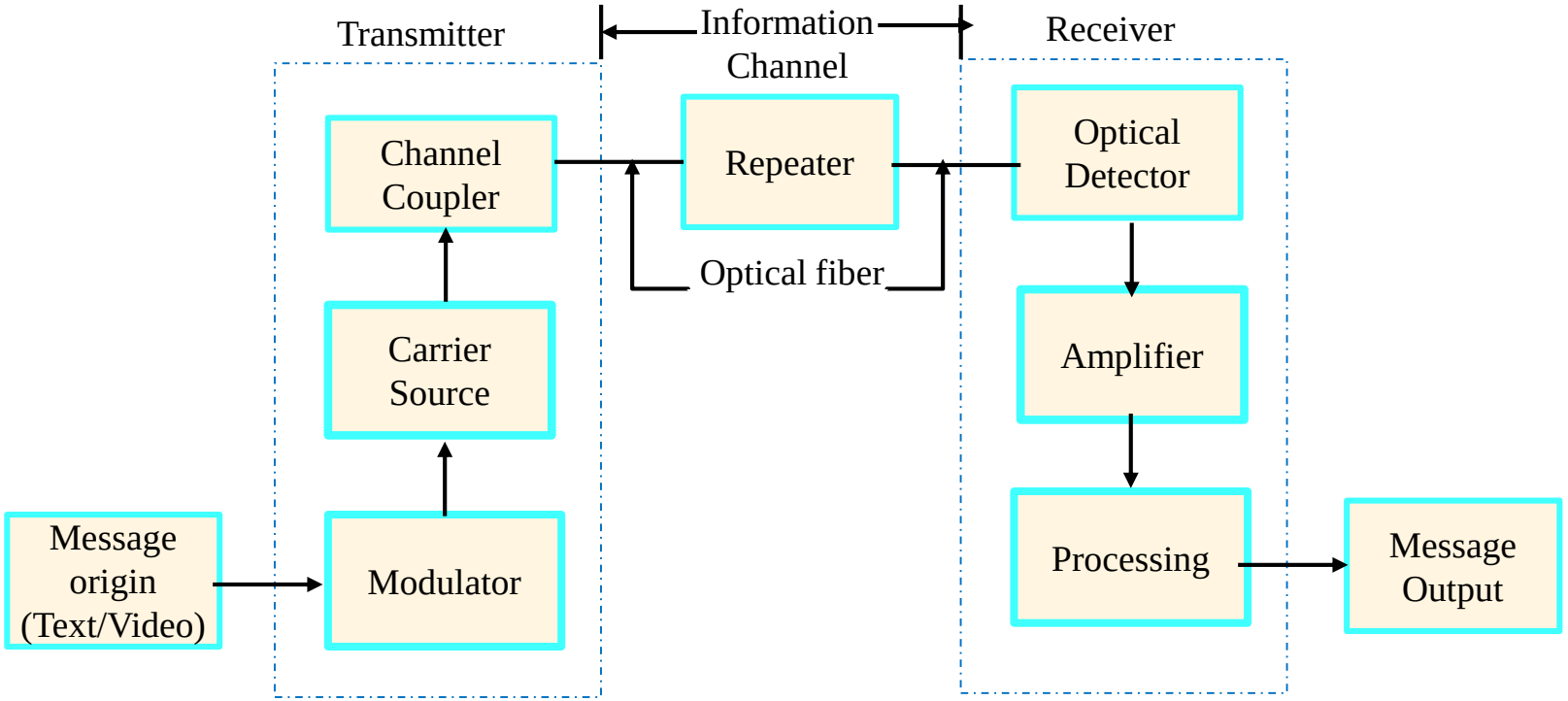



Fig. Block diagram of OFC systems

# General OFC System

## Message origin:

- It is transducer convert non-electrical message to electrical signal.
- e.g., microphone, video camera

## Modulator:

- It converts the electrical message into proper format.
- It impresses this signal onto the wave generated by the carrier source.
- Two distinct categories of modulation are used i.e. analog modulation and digital modulation.

## Carrier source:

- Carrier source generates the wave on which the information is transmitted
- For fiber optic system, a laser diode (LD) or a light emitting diode (LED) is used.

# General OFC System

## Channel coupler:

- Coupler feeds the power into the information channel
- Generally, lens are used as a channel coupler.
- The lens collimates the light emitted by the source and directing this light towards the receiver through fiber.

## Information channel:

- The information channel is the path between the transmitter and receiver.
- A glass or plastic fiber is used as a information channel.
- The channel attenuation should be low and large light acceptance angle.
- Repeater are used to regenerate weak and distorted signal.
- Another property of channel is propagation time. The signal in a channel contains different optic frequencies and divide its power along several ray paths, that cause pulse spreading.

# General OFC System

## Optical detector:

- In the fiber system the optic wave is converted into an electric current by a photodetector.
- The important properties of photodetectors are small size, economy, long life, low power consumption, high sensitivity to optic signals and fast response to quick variations in the optic power.

## Signal processing:

- Signal processing includes filtering, amplification.
- Proper filtering maximizes the ratio of signal to unwanted power.
- For a digital system, the BER should be small.

Bit Error Rate

## Message output:

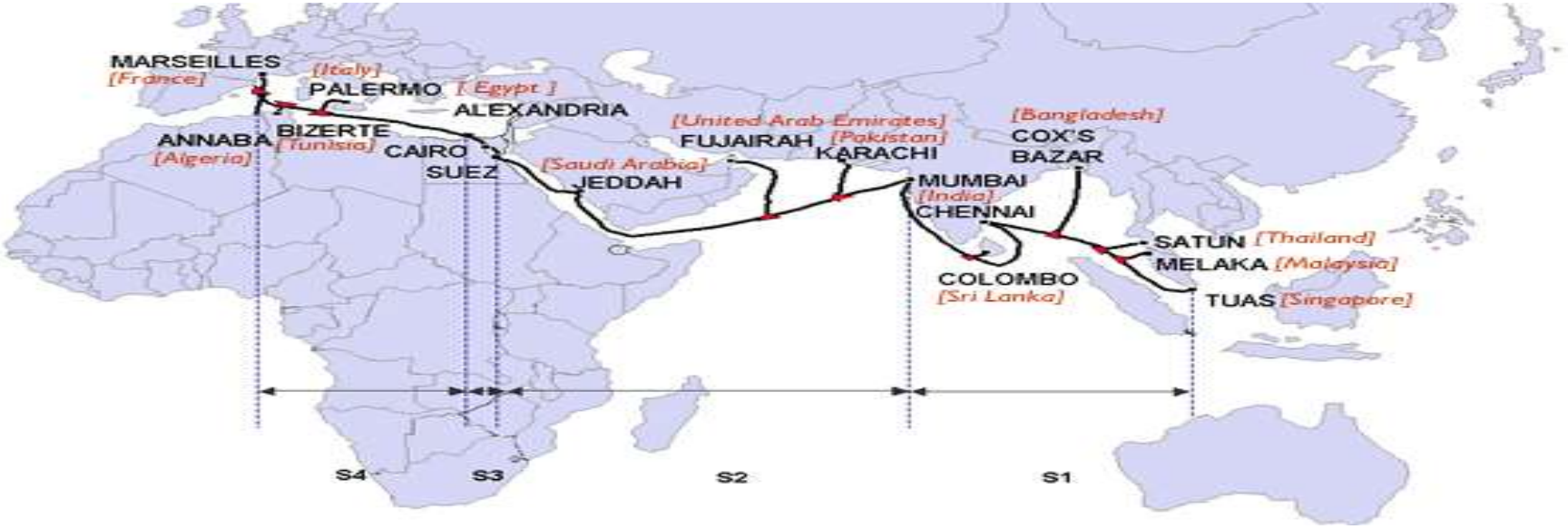
- The electrical form of the message emerging from the signal processor is transformed into a sound wave or visual image.
- Sometimes these signals are directly usable when computers or other machines are connected through a fiber system.

# Elements

- Source & Modulators
  - LED, LD etc.
- Multiplexers & Demultiplexers
  - WDM, DWDM etc.
- Amplifiers or Repeaters
- Connectors, Couplers, Isolators, etc.
- Detectors & Demodulators
  - PD, APD, PIN etc.
- End Mile Networks

# SEA-ME-WE

**South East Asia–Middle East–Western Europe 4** (**SEA-ME-WE 4**) is an optical fibre submarine communications cable system that carries telecommunications between Singapore, Malaysia, Thailand, Bangladesh, India, Sri Lanka, Pakistan, United Arab Emirates, Saudi Arabia, Sudan, Egypt, Italy, Tunisia, Algeria and France.[1] It is intended to be a complement to, rather than a replacement for, the SEA-ME-WE 3 cable. The cable is approximately 18,800 kilometres long, and provides the primary Internet backbone between South East Asia, the Indian subcontinent, the Middle East and Europe



# History of SEA-ME-WE

- South-East Asia - Middle East - Western Europe
- 1986, the first cable of the SEA-ME-WE series (SEA-ME-WE 1) was constructed using a coaxial cable designed to transmit telephone signals between Europe, the Middle East and Southeast Asia.
- It was the longest telephone cable in the world spanning 13,585 km, and was the first cable to be laid in the Indian Ocean.
- The maximum bandwidth of the cable was calculated to be 25 MHz
- Due to higher demands of broadband and data transfer, leading to the construction of the SEA-ME-WE 2, 3 and 4 projects.

# SEA-ME-WE 2

# SEA-ME-WE 2

- South-East Asia - Middle East - Western Europe 2.
- The SEA-ME-WE 2 project was installed on October 18th, 1994.
- A  fiber optic cable system, spanning 18,751 km, linking 13 countries and three continents.
- Peak capacity of the cable was 560 Mbps, which is approximately equal to the capacity of fifteen SEA-ME-WE 1 cables.

# SEA-ME-WE 2 Route



# Introduction To SEA-ME-WE 3

**SEA-ME-WE 3**

South-East Asia - Middle East - Western Europe 3.

- optical submarine telecommunication cable link and is the longest in the world stretches from North Germany to Australia and Japan.

- completed in late 2000.

# Introduction …

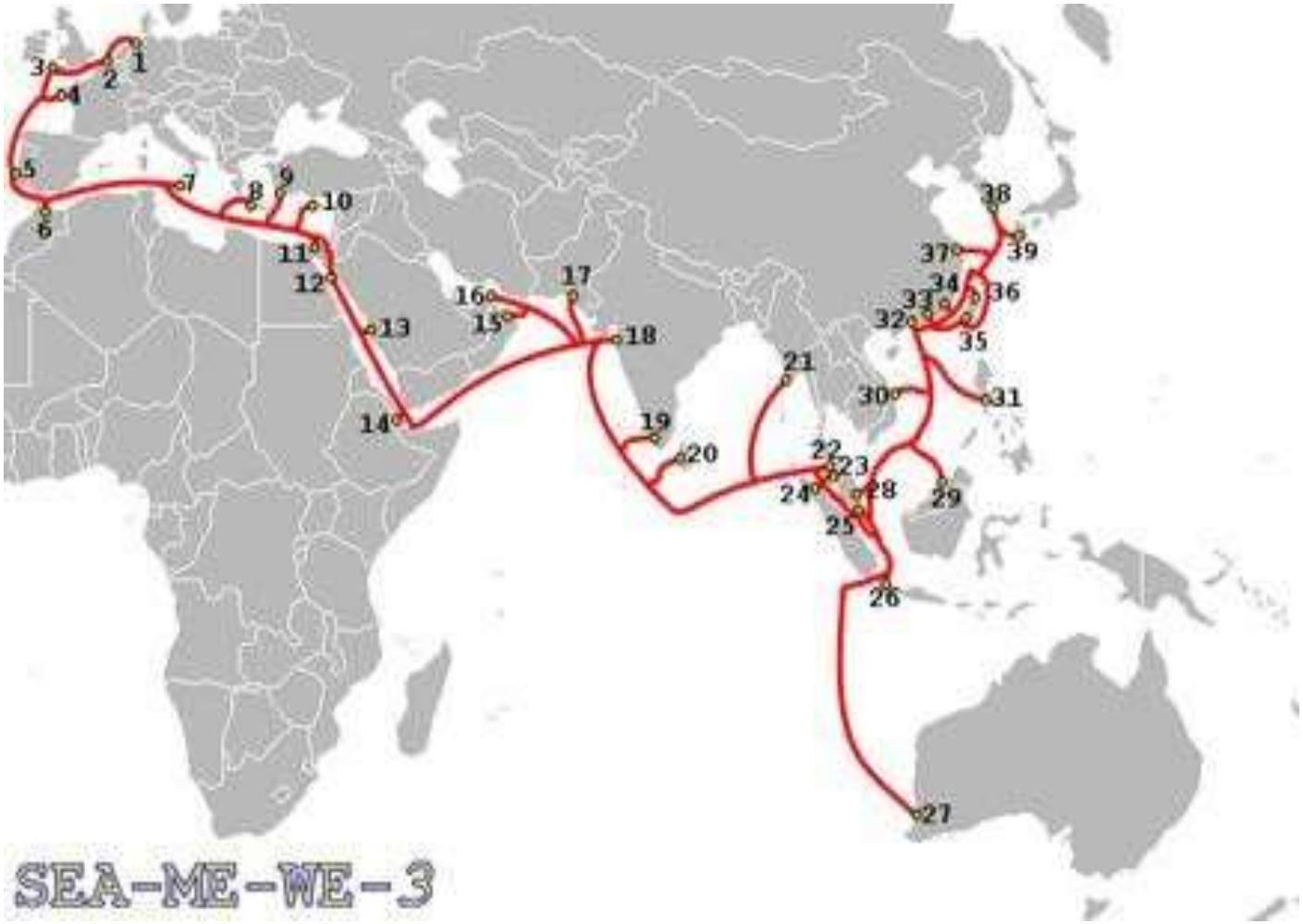
- led by France Telecom and China Telecom, and is administered by SingTel.
- telecommunications operator owned by the Government of Singapore.
- Total length 39,000 kilometers (24,000 mi).

# Technology used in SEA-ME-WE 3

- Wavelength Division Multiplexing (*WDM*) technology is for transmission to increase capacity and enhance the quality of the signal, especially over long distances.

- **WDM** is a technology which multiplexes a number of optical carrier signals onto a single optical fiber by using different wavelengths (i.e. colors) of laser light.

# Route and landing points SEA-ME-WE 3



# Landing points of SEA-ME-WE 3

- 39 landing points:

1. Norden, Germany
2. Oostende, Belgium
3. Goonhilly, England, UK
4. Penmarch, France
5. Sesimbra, Portugal
6. Tetuan, Morocco
7. Mazara del Vallo, Italy
8. Chania, Greece
9. Marmaris, Turkey
10. Yeroskipou, Cyprus
11. Alexandria, Egypt
12. Suez, Egypt
13. Jeddah, Saudi Arabia
14. Djibouti, Djibouti
15. Muscat, Oman
16. Fujairah, United Arab Emirates
17. Karachi, Pakistan
18. Mumbai, India
19. Cochin, India
20. Mount Lavinia, Sri Lanka

# Landing points…

21. Pyapon, Myanmar
22. Satun, Thailand
23. Penang, Malaysia (Where it meets the SAFE and the FLAG cables.)
24. Medan, Indonesia
25. Tuas, Singapore
26. Jakarta, Indonesia
27. Perth, Australia
28. Mersing, Malaysia
29. Tungku, Brunei
30. Da Nang, Vietnam
31. Batangas, Philippines
32. Taipa, Macau
33. Deep Water Bay, Hong Kong
34. Shantou, China
35. Fangshan, Taiwan
36. Toucheng, Taiwan
37. Shanghai, China
38. Keoje, South Korea
39. Okinawa, Japan

# SEA-ME-WE 4

# SEA-ME-WE 4

- South East Asia-Middle East-West Europe 4 (SEA-ME-WE 4)

- It is the fourth project in the SEA-ME-WE series. On 27th March 2004.

# SEA-ME-WE 4 …

- an optical fiber submarine communications cable.
- carries telecommunications between Singapore, Malaysia, Thailand, Bangladesh, India, Sri Lanka, Pakistan, United Arab Emirates, Saudi Arabia, Sudan, Egypt, Italy, Tunisia, Algeria and France.

# History of SEA-ME-WE 4

- SEA-ME-WE 4 cable system was developed by a consortium of 16 telecommunications companies which agreed to construct the project on 27 March 2004.
- Construction of the system was carried out by Alcatel Submarine Networks
- project was completed on 13 December 2005 with a cost estimate of US$500 million

# Network topology used in SEA-ME-WE 4

- The SEA-ME-WE 4 system is divided into four segments with seventeen landing points:

**Segments**

S1 - Tuas to Mumbai

S2 - Mumbai to Suez

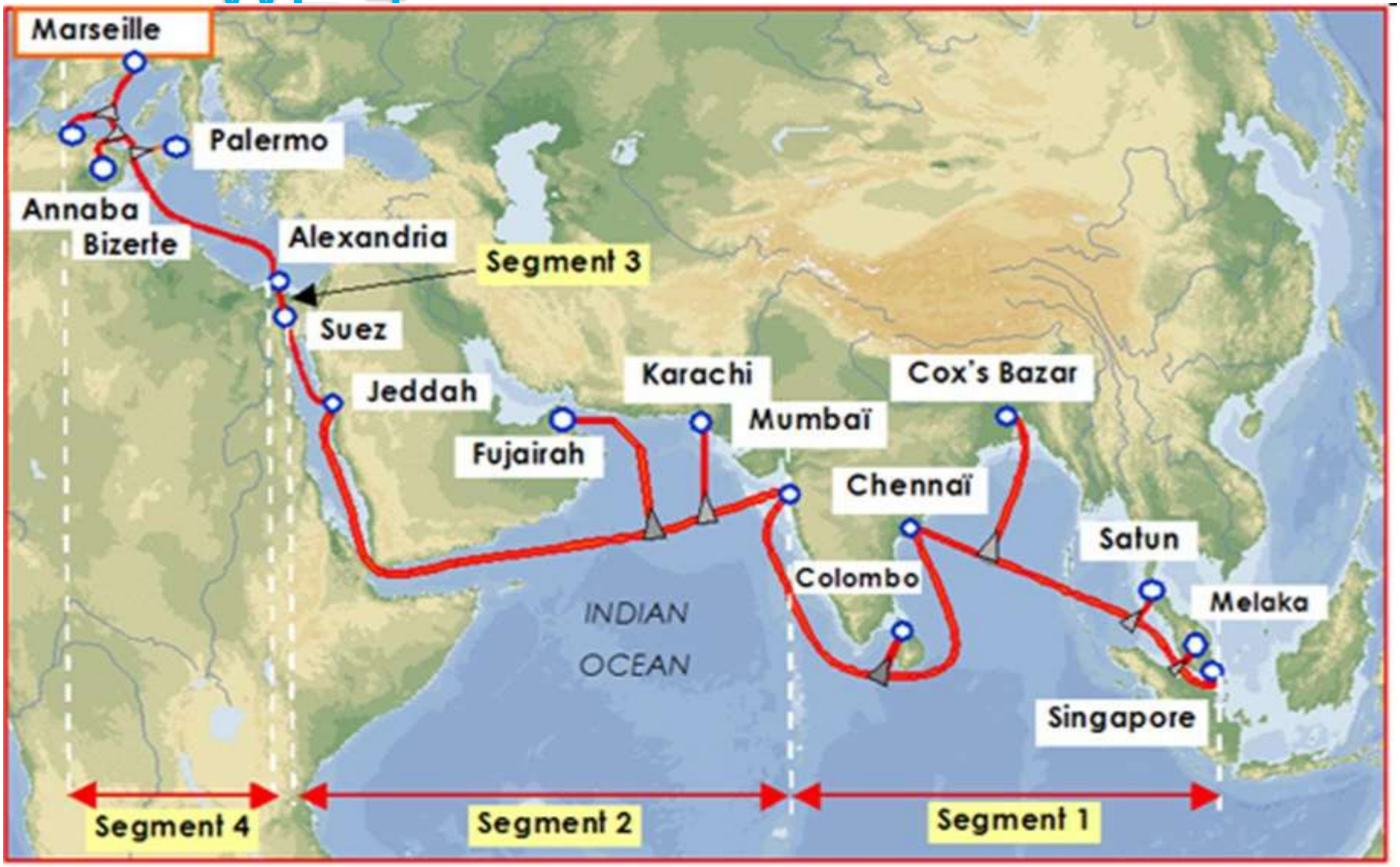
S3 - Suez to Cairo

S4 - Cairo to Marseille

# Countries linked through SEA-ME-WE 4

- Bangladesh
- India
- Pakistan
- Sri-Lanka
- Thailand
- Malaysia
- Singapore
- United Arab Emirates
- Sudan
- Egypt
- Italy
- Tunisia
- Algeria
- France
- Saudi Arabia

# Connecting Point of SEA-ME-WE 4



# Length of SEA-ME-WE 4

- The cable is approximately 18,800 kilometers long.
- provides the primary Internet backbone between South East Asia, the Indian subcontinent, the Middle East and Europe.

# Technology used in SEA-ME-WE 4

- Uses wavelength-division multiplexing WDM, allowing increased communications capacity per fiber.
- It also facilitates bidirectional communication within a single fiber.

# Advantages of SEA-ME-WE 4

- SEA-ME-WE-4 is used to carry telephone, internet, multimedia, broadband data.
- SEA-ME-WE 4 has a faster rate of data transmission at 1.28 Tbit/s against SEA-ME-WE 3's 10 Gbit/s
- SEA-ME-WE 3 provides connectivity to a greater number of countries over a greater distance, but SEA-ME-WE 4 provides far higher data transmission speeds intended to accommodate increasing demand for high-speed internet access in developing countries

# Comparison Between SEA-ME-WE 1 and SEA-ME-WE 2

## SEA-ME-WE 1

- coaxial cable designed to transmit telephone signals.
- length13,585 km.
- maximum bandwidth of the cable was 25 MHz

## SEA-ME-WE 2

- fiber optic cable system.
- 18,751 km, linking 13 countries.
- capacity of the cable was 560 Mbps.

# Comparison Between SEA-ME-WE 3 and SEA-ME-WE 4

SEA-ME-WE 3

- SEA-ME-WE 3 cable was completed with a capacity of 2.5 Gbps, which was later upgraded to 10 Gbps in 2002.
- Total length 38,000 km.

**SEA-ME-WE 4**

- Total length 20,000 km.
- Capacity of 1.28 Tbps,

SEA-ME-WE 5




| | SEA-ME-WE | SEA-ME-WE 2 | SEA-ME-WE 3 | SEA-ME-WE 4 |
|---|---|---|---|---|
| **Commissioned in** | June 1985 (Decommissioned June 1999) | October 1994 (Decommissioned October 2006) | August 1999 | November 2005 |
| **Capacity/Length** | 12MHz 13,500km | 2x560Mbps 18,000km | 8x2.5Gbps 39,000km | 64x2x10Gbps 20,000km |
| **No. of Owners Total Investment** | 22 USD800M | 52 USD800M | 92 USD1500M | 16 USD500M |
| **Technology** | Analog/Copper | PDH/Optical | SDH/WDM/Optical | SDH/DWDM/Optical |

SEA-ME-WE 5

- Matchless, **PoP to PoP, multi-regional data superhighway**
- **20.000 km** submarine cable system connecting Southeast Asia to Western Europe
- Further enhancing the **diversity** and **resilience** to the heavily loaded Asia to Europe route
- **Latest 100Gbps technology** enabling initial system capacity of **24 Tbps**
- **High level of flexibility** for consortium members in utilizing their capacity entitlement
- **Lowest latency**
- Ready for service: **Q4 2016**

SEA-ME-WE 5


SEA-ME-WE 5 route
Planned RFS: Q4 2016
Technology utilised: 100Gbps DWDM
Design capacity: 24 Tbps
Length of the cable: 20 000 km
Toulon
Catania
Marmaris
Abu Talat
Zafarana
Yanbu
Djibouti
Kalba
Qalhat
Karachi
Mumbai
Kuakata
Pathein
Matara
Melaka
Medan
Dumai
Tuas

SEA-ME-WE 5

- Established by **industry leading operators** with extensive experience in submarine cable projects.
- Consortium's **proven track record** of delivering results from successful completion of previous SEA-ME-WE projects.
- **Financially stable** - fully secured financing.
- **Strong commitment** by all participating parties.
- **Leading-edge technical know-how** of its consortium members.
- **Open system** with free access to carrier neutral, international PoPs belonging to the core of SEA-ME-WE 5.
- **Solid design** and well-founded project plan which has been followed without any deviation to date.
- The spans of the system are optimized for **future upgradability**.

SEA-ME-WE 5

- The core system includes **fully protected terrestrial extensions** to major carrier-neutral PoPs. The list of the main PoPs which are connected to SEA-ME-WE 5:
  - **Marseille**: Interxion (formerly Net Center) and/or Jaguar
  - **Palermo:** New carrier neutral Sicily Hub PoP
  - **Singapore**: Equinix, Global Switch
- SEA-ME-WE 5 will also **secure outstanding cross connection possibilities** with other submarine cables in **Egypt, Saudi Arabia, Djibouti and Sri Lanka.**
- **Marmaris CLS in Turkey** will also be opened without extra costs for members to increase the options for terminating capacities in Europe.

SEA-ME-WE 5


Marseille
Toulon
Palermo
Catania
Marseille POPs & Palermo POP


Medan
Melaka
Singapore
Dumai
Tuas
Singapore POPs

# SEA-ME-WE 5

| | | | | | |
|---|---|---|---|---|---|
| Bangladesh Submarine Cable Company Limited (BSCCL) | Kuakata CLS | Myanmar Post and Telecom (MPT) | Pathein CLS | Sri Lanka Telecom PLC (SLT) | Matara CLS |
| China Mobile International Limited (CMI) | non-landing Party | Orange | Toulon CLS, Marseille PoP | Telecom Egypt | Zafarana CLS<br>Abu Talat CLS |
| China Telecom Global Limited (CTG) | non-landing Party | Ooredoo Qatar/Oman | Qatar<br>Qalhat CLS | Telekom Malaysia Berhad (TM) | Melaka CLS |
| China United Network Communications Group Company Limited (CU) | non-landing Party | PT Telekomunikasi Indonesia International (Telin) | Dumai CLS<br>Medan CLS | Telecom Italia Sparkle (TI Sparkle) | Catania CLS<br>Palermo PoP |
| Djibouti Telecom | Djibouti CLS | Saudi Telecom Company (STC) | Yanbu CLS | Türk Telekom International (TTI) | Marmaris CLS |
| Emirates Integrated Telecommunications Company (Du) | Kalba CLS | Singapore Telecommunications Limited (Singtel) | Tuas CLS<br>Singapore PoPs | TWA | Karachi CLS |

SEA-ME-WE 5

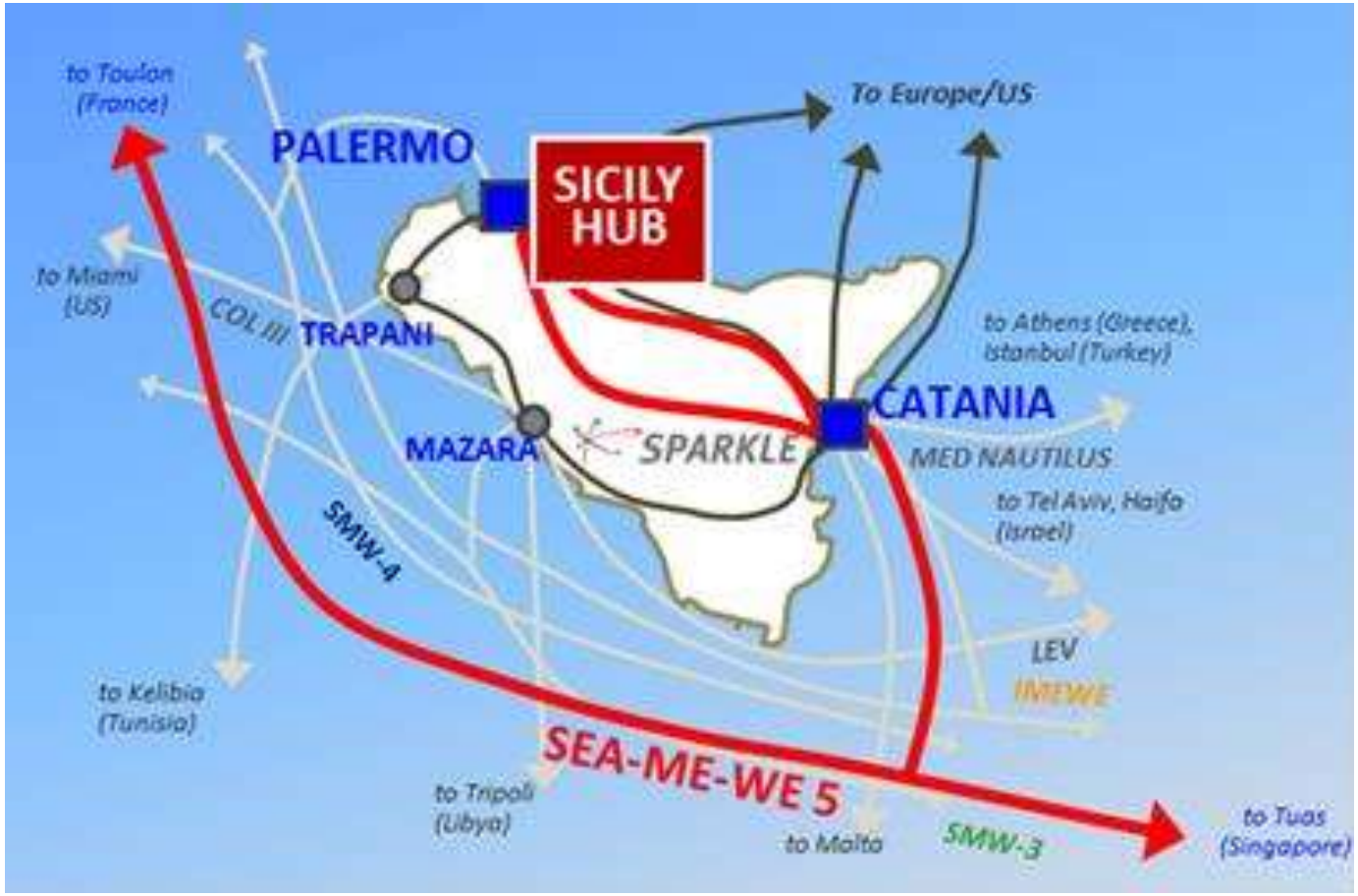

SEA-ME-WE 5 route
Planned RFS: Q4 2016
Technology utilised: 100Gbps DWDM
Design capacity: 24 Tbps
Length of the cable: 20 000 km
Toulon
Catania
Marmaris
Abu Talat
Zafarana
Yanbu
Djibouti
Kalba
Qalhat
Karachi
Mumbai
Kuakata
Pathein
Matara
Melaka
Medan
Dumai
Tuas

# SEA-ME-WE 5


SMW5 cable station to be connected with Jeddah CLS the strategic regional hub as a key link between the east and the west
TURKEY
SYRIA
JORDAN
OTHER COUNTRIES
IRAQ
EGYPT
JADI
SAIT
RED SEA
KUWAIT
QATAR
U.A.E.
SWM4
EIG
MENA
AL-MUJAMAH
JEDDAH
AL-SALAMAH
BAHRAIN
QATAR
SAUDI ARABIA
SAS1
IMEWE
EASSY
FLAG
SMW3
SUDAN
UAE
YEMEN
OMAN
STC
الاتصالات السعودية

SEA-ME-WE 5

Catania CLS
and
Palermo PoP
in ITALY



SEA-ME-WE 5


SEA-ME-WE 5 route
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Pathein
Matara
Melaka
Medan
Dumai
Tuas

# Advantages of Optical Fiber

Advantages of OFC:

- Extremely high bandwidth:
- Longer distance: Typically, it is less than 1dB/km.
- Immune to cross talk
- Resistance to electromagnetic interference
- Low security risk
- Small size
- Light weight
- Greater flexibility
- Easy accommodate increasing bandwidth

# Disadvantages of Optical Fiber

disadvantages of OFC:

- High initial cost
- Maintenance and repairing cost
- Jointing and test procedures
- Tensile stress
- Fiber losses

# Computing power levels (decibels)

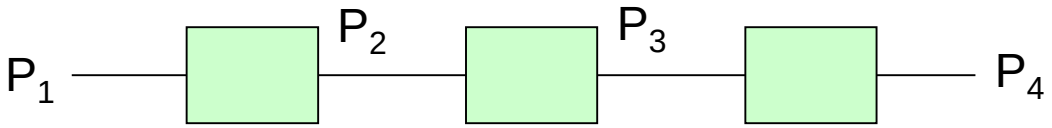
- To design an optical fiber communication systems, it is important to account the optical power.
- This power measurement helps to ensure that incident wave on the detector has sufficient strength to be clearly and correctly recognized.
- In other instances the received power may even be too large for the receiver.
- The designer must be certain this does not occur.

If the power is $P_1$ watts at one point in the system and $P_2$ watts at some point farther along the link, then $P_2/P_1$ is the fraction of the power transmitted between the two locations

$$dB = 10 \log_{10} P_2/P_1 \text{ or, } P_2 = P_1 \, 10^{dB/10}$$

# Decibels (dB) Calculation

If we consider power levels in a cascaded system of the following figure:




The output power is determined by:

$$\frac{P_4}{P_1} = \frac{P_4}{P_3} \times \frac{P_3}{P_2} \times \frac{P_2}{P_1}$$

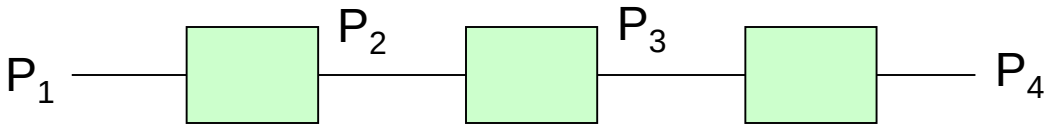
The logarithmic scale is used to ease with which total change in power level is found when several elements are cascaded:

$$dB = 10\log_{10}\frac{P_4}{P_1} = 10\log_{10}\left(\frac{P_4}{P_3} \times \frac{P_3}{P_2} \times \frac{P_2}{P_1}\right)$$

$$= 10\log_{10}\frac{P_4}{P_3} = 10\log_{10}\frac{P_3}{P_2} = 10\log_{10}\frac{P_2}{P_1}$$

# Example

Suppose that the three elements in the following figure have losses -11,-6 and -3 dB respectively. Find the total loss of the combination. Find the output power if the input power is 5mW.




Solution: The total loss is =-11-6-3=-20dB

$$dB=10\log_{10}\frac{P_o}{P_{in}}$$

$$-20=10\log_{10}\frac{P_o}{P_{in}}$$

$$\frac{P_o}{P_{in}}=10^{\frac{-20}{10}}=10^{-2}=0.01$$

$$P_0=0.01\times5=0.05\ \text{mW}$$

# Example

A system has -23 dB of loss. Compute its efficiency.

✅ **Problem 1: Efficiency from dB loss**

Given loss = **–23 dB**

Formula:

$$\text{Efficiency} = 10^{\frac{\text{dB}}{10}}$$

$$\eta = 10^{\frac{-23}{10}} = 10^{-2.3}$$

$$\eta \approx 0.00501$$

👉 **Efficiency ≈ 0.005 (or 0.5%)**

A light emitting diode radiates 2mW. Compute the dBm value of this radiated power. The power travels through a group of components having a combined loss of 23 dB. Compute the output power.

✅ **Problem 2: LED Power (dBm and Output Power)**

**Step 1: Convert 2 mW to dBm**

Formula:

$$P(dBm) = 10\log_{10}(P_{mW})$$

$$P = 10\log_{10}(2)$$

$$P \approx 10 \times 0.301 = 3.01 \text{ dBm}$$

👉 **Input Power ≈ 3.01 dBm**

**Step 2: Apply loss (23 dB)**

$$P_{out} = 3.01 - 23 = -19.99 \approx -20 \text{ dBm}$$

👉 **Output Power ≈ –20 dBm**

**Step 3: Convert back to mW**

Formula:

$$P(mW) = 10^{\frac{dBm}{10}}$$

$$P = 10^{\frac{-20}{10}} = 10^{-2}$$

$$P = 0.01 \text{ mW}$$

👉 **Output Power ≈ 0.01 mW (or 10 μW)**